

Summary of Algebraic Formulas and Theorems

Part 1 : Logic / Logique

1.1 Propositional Logic / Logique des propositions

- Definition of proposition / Définition de proposition : P true or false
- Truth values / Valeurs de vérité : 1 (true), 0 (false)
- Truth tables / Tables de vérité :

P	Q	$P \wedge Q$	P	Q	$P \vee Q$
0	0	0	0	0	0
0	1	0	0	1	1
1	0	0	1	0	1
1	1	1	1	1	1

- Negation / Négation : $\overline{P}$, $\neg P$

P	$\overline{P}$
0	1
1	0

- Double negation / Double négation :

$$\overline{\overline{P}} \iff P$$

- Implication / Implication :

$$P \implies Q \iff \overline{P} \vee Q$$

- Contrapositive / Contraposée :

$$(P \implies Q) \iff (\overline{Q} \implies \overline{P})$$

- Reciprocal / Réciproque :

$$(Q \implies P)$$

- Equivalence / Équivalence :

$$P \iff Q \iff (P \implies Q) \wedge (Q \implies P)$$

- De Morgan's laws / Lois de De Morgan :

$$\begin{aligned}\overline{P \wedge Q} &\iff \overline{P} \vee \overline{Q} \\ \overline{P \vee Q} &\iff \overline{P} \wedge \overline{Q}\end{aligned}$$

— **Logical properties / Propriétés logiques :**

$$P \wedge P \iff P \quad (\text{Idempotence})$$

$$P \vee P \iff P$$

$$P \wedge \bar{P} \iff 0 \quad (\text{Contradiction})$$

$$P \vee \bar{P} \iff 1 \quad (\text{Tertium non datur})$$

$$(P \wedge Q) \wedge R \iff P \wedge (Q \wedge R) \quad (\text{Associativity})$$

$$(P \vee Q) \vee R \iff P \vee (Q \vee R)$$

$$P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R) \quad (\text{Distributivity})$$

$$P \vee (Q \wedge R) \iff (P \vee Q) \wedge (P \vee R)$$

$$(P \implies Q) \wedge (Q \implies R) \implies (P \implies R) \quad (\text{Transitivity})$$

1.2 Quantifiers / Quantificateurs

— **Universal quantifier / Quantificateur universel :**

$$\forall x \in E, P(x)$$

— **Existential quantifier / Quantificateur existentiel :**

$$\exists x \in E, P(x)$$

— **Unique existence / Existence unique :**

$$\exists! x \in E, P(x)$$

— **Negation of quantifiers / Négation des quantificateurs :**

$$\neg(\forall x, P(x)) \iff \exists x, \neg P(x)$$

$$\neg(\exists x, P(x)) \iff \forall x, \neg P(x)$$

Part 2 : Set Theory / Théorie des ensembles

2.1 Basic Definitions / Définitions de base

— **Set / Ensemble :** Collection of objects

— **Element / Élément :** $x \in E, x \notin E$

— **Empty set / Ensemble vide :** $\emptyset, \text{Card}(\emptyset) = 0$

— **Singleton / Singleton :** $\text{Card}(A) = 1$

— **Cardinal / Cardinal :** $\text{Card}(E)$ (finite) or ∞ (infinite)

— **Power set / Ensemble des parties :** $\mathcal{P}(E)$

$$\text{If } \text{Card}(E) = n, \text{ then } \text{Card}(\mathcal{P}(E)) = 2^n$$

— **Subset / Sous-ensemble :**

$$A \subseteq B \iff \forall x(x \in A \implies x \in B)$$

— **Proper subset / Sous-ensemble propre :**

$$A \subsetneq B \iff (A \subseteq B) \wedge (A \neq B)$$

— **Equality of sets / Égalité d'ensembles :**

$$A = B \iff (A \subseteq B) \wedge (B \subseteq A)$$

2.2 Set Operations / Opérations sur les ensembles

— Intersection / Intersection :

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

— Union / Réunion :

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

— Difference / Différence :

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}$$

— Complement / Complémentaire :

$$\complement_E A = E \setminus A = \{x \in E \mid x \notin A\}$$

— Cartesian product / Produit cartésien :

$$A \times B = \{(x, y) \mid x \in A, y \in B\}$$

$$\text{Card}(A \times B) = \text{Card}(A) \times \text{Card}(B)$$

— Properties of operations / Propriétés des opérations :

$$A \cap A = A, \quad A \cup A = A \quad (\text{Idempotence})$$

$$A \cap B = B \cap A, \quad A \cup B = B \cup A \quad (\text{Commutativity})$$

$$(A \cap B) \cap C = A \cap (B \cap C) \quad (\text{Associativity})$$

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \quad (\text{Distributivity})$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap \emptyset = \emptyset, \quad A \cup \emptyset = A$$

$$A \cap E = A, \quad A \cup E = E \quad (\text{si } A \subseteq E)$$

$$\complement_E(\complement_E A) = A$$

$$\complement_E(A \cap B) = \complement_E A \cup \complement_E B \quad (\text{De Morgan})$$

$$\complement_E(A \cup B) = \complement_E A \cap \complement_E B$$

$$A \subseteq B \iff \complement_E B \subseteq \complement_E A$$

— Partition / Partition :

$\mathcal{F} \subset \mathcal{P}(E)$ est une partition si :

$$1. \bigcup_{A \in \mathcal{F}} A = E$$

$$2. \forall A, B \in \mathcal{F}, A \neq B \implies A \cap B = \emptyset$$

2.3 Functions / Applications

— Function definition / Définition d'application :

$$f : E \rightarrow F, \quad \forall x \in E, \exists! y \in F, y = f(x)$$

— Identity function / Application identité :

$$\text{Id}_E : E \rightarrow E, \quad \text{Id}_E(x) = x$$

— **Constant function / Application constante :**

$$\forall x \in E, f(x) = c$$

— **Restriction / Restriction :**

$$f|_A : A \rightarrow F, \quad f|_A(x) = f(x)$$

— **Extension / Prolongement :** $g : G \rightarrow F$ with $E \subseteq G, g|_E = f$

— **Composition / Composition :**

$$(g \circ f)(x) = g(f(x))$$

$$h \circ (g \circ f) = (h \circ g) \circ f \quad (\text{Associativity})$$

— **Image and preimage / Image et image réciproque :**

$$\begin{aligned} f(A) &= \{f(x) \mid x \in A\} \\ f^{-1}(B) &= \{x \in E \mid f(x) \in B\} \end{aligned}$$

— **Properties / Propriétés :**

$$\begin{aligned} f(A \cup B) &= f(A) \cup f(B) \\ f(A \cap B) &\subseteq f(A) \cap f(B) \\ f^{-1}(A \cup B) &= f^{-1}(A) \cup f^{-1}(B) \\ f^{-1}(A \cap B) &= f^{-1}(A) \cap f^{-1}(B) \\ f^{-1}(\mathcal{C}_F A) &= \mathcal{C}_E f^{-1}(A) \end{aligned}$$

— **Injectivity / Injectivité :**

$$f \text{ injective} \iff \forall x_1, x_2 \in E, f(x_1) = f(x_2) \implies x_1 = x_2$$

— **Surjectivity / Surjectivité :**

$$f \text{ surjective} \iff \forall y \in F, \exists x \in E, f(x) = y$$

— **Bijectivity / Bijectivité :**

$$f \text{ bijective} \iff f \text{ injective et surjective}$$

— **Inverse function / Fonction inverse :**

$$f^{-1} : F \rightarrow E, \quad f^{-1}(y) = x \iff f(x) = y$$

$$f^{-1} \circ f = Id_E, \quad f \circ f^{-1} = Id_F$$

— **Properties of bijections / Propriétés des bijections :**

$$(f^{-1})^{-1} = f$$

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

If f and g injective, then $g \circ f$ injective

If f and g surjective, then $g \circ f$ surjective

If $g \circ f$ injective, then f injective

If $g \circ f$ surjective, then g surjective

Part 3 : Relations / Relations

3.1 Equivalence Relations / Relations d'équivalence

— **Definition / Définition** : R is equivalence if :

1. Reflexive / Réflexive : $\forall x \in E, xRx$
2. Symmetric / Symétrique : $\forall x, y \in E, xRy \implies yRx$
3. Transitive / Transitive : $\forall x, y, z \in E, (xRy \wedge yRz) \implies xRz$

— **Equivalence class / Classe d'équivalence** :

$$\dot{x} = [x] = \{y \in E \mid xRy\}$$

— **Quotient set / Ensemble quotient** :

$$E/R = \{\dot{x} \mid x \in E\}$$

— **Canonical surjection / Surjection canonique** :

$$s : E \rightarrow E/R, \quad s(x) = \dot{x}$$

— **Properties / Propriétés** :

$$\dot{x} \cap \dot{y} = \emptyset \quad \text{or} \quad \dot{x} = \dot{y}$$

E/R is a partition of E

$$\bigcup_{x \in E} \dot{x} = E$$

3.2 Order Relations / Relations d'ordre

— **Definition / Définition** : $\preceq$ is order if :

1. Reflexive / Réflexive : $\forall x \in E, x \preceq x$
2. Antisymmetric / Antisymétrique : $\forall x, y \in E, (x \preceq y \wedge y \preceq x) \implies x = y$
3. Transitive / Transitive : $\forall x, y, z \in E, (x \preceq y \wedge y \preceq z) \implies x \preceq z$

— **Total order / Ordre total** :

$$\forall x, y \in E, x \preceq y \quad \text{or} \quad y \preceq x$$

— **Partial order / Ordre partiel** : Not total order

— **Minimum element / Plus petit élément** :

$$m \in A, \quad \forall y \in A, m \preceq y$$

— **Maximum element / Plus grand élément** :

$$M \in A, \quad \forall y \in A, y \preceq M$$

— **Minimal element / Élément minimal** :

$$m \in A, \quad \forall y \in A, y \preceq m \implies y = m$$

— **Maximal element / Élément maximal** :

$$M \in A, \quad \forall y \in A, M \preceq y \implies y = M$$

— **Lower bound / Minorant :**

$$m \in E, \quad \forall x \in A, m \preceq x$$

— **Upper bound / Majorant :**

$$M \in E, \quad \forall x \in A, x \preceq M$$

— **Infimum / Borne inférieure :**

$$\inf A = \text{greatest lower bound}$$

— **Supremum / Borne supérieure :**

$$\sup A = \text{least upper bound}$$

— **Bounded set / Ensemble borné :** Has both lower and upper bounds

— **Properties / Propriétés :**

$$\text{Minimum} \implies \text{Minimal (unique)}$$

$$\text{Maximum} \implies \text{Maximal (unique)}$$

$$\inf A \text{ unique if exists}$$

$$\sup A \text{ unique if exists}$$

$$\sup(A \cup B) = \max(\sup A, \sup B)$$

$$\inf(A \cup B) = \min(\inf A, \inf B)$$

Part 4 : Algebraic Structures / Structures algébriques

4.1 Internal Composition Laws / Lois de composition internes

— **Definition / Définition :** $*$: $E \times E \rightarrow E$

— **Stability / Stabilité :**

$$\forall a, b \in F \subseteq E, a * b \in F$$

— **Commutativity / Commutativité :**

$$\forall a, b \in E, a * b = b * a$$

— **Associativity / Associativité :**

$$\forall a, b, c \in E, (a * b) * c = a * (b * c)$$

— **Distributivity / Distributivité :**

$$\forall a, b, c \in E, a * (b \cdot c) = (a * b) \cdot (a * c)$$

$$\forall a, b, c \in E, (b \cdot c) * a = (b * a) \cdot (c * a)$$

— **Neutral element / Élément neutre :**

$$\exists e \in E, \forall a \in E, e * a = a * e = a$$

Neutral element is unique

If e' right neutral and e'' left neutral, then $e' = e''$

— **Inverse element / Éléments inverse :**

$$\forall a \in E, \exists a^{-1} \in E, a * a^{-1} = a^{-1} * a = e$$

If $*$ associative, inverse is unique

$$e^{-1} = e$$

$$(a^{-1})^{-1} = a$$

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

— **Regular element / Éléments régulier :**

$$\forall b, c \in E, a * b = a * c \implies b = c \quad (\text{left regular})$$

$$\forall b, c \in E, b * a = c * a \implies b = c \quad (\text{right regular})$$

$$\text{Invertible} \implies \text{Regular}$$

— **Examples / Exemples :**

$(P(E), \cap)$: neutral E , associative, commutative

$(P(E), \cup)$: neutral $\emptyset$, associative, commutative

$\cap$ distributive over $\cup$ and vice versa

4.2 Groups / Groupes

— **Definition / Définition :** $(G, *)$ is a group if :

1. $*$ is associative
2. $\exists e \in G$ neutral element
3. $\forall a \in G, \exists a^{-1} \in G$

— **Abelian group / Groupe abélien :** $*$ is commutative

— **Subgroup / Sous-groupe :** $H \subseteq G$ is subgroup if :

$$H \neq \emptyset$$

$$\forall a, b \in H, a * b^{-1} \in H$$

or equivalently :

$$e \in H$$

$$\forall a, b \in H, a * b^{-1} \in H$$

— **Trivial subgroup / Sous-groupe trivial :** $\{e\}, G$

— **Proper subgroup / Sous-groupe propre :** $H \neq \{e\}$ and $H \neq G$

— **Examples / Exemples :**

$(\mathbb{Z}, +)$: abelian group

$(\mathbb{R}, +), (\mathbb{Q}, +), (\mathbb{C}, +)$

$(\mathbb{R}^*, \cdot), (\mathbb{Q}^*, \cdot), (\mathbb{C}^*, \cdot)$

$n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ subgroup of $(\mathbb{Z}, +)$

— **Group properties / Propriétés des groupes :**

$$\begin{aligned}
 (a^{-1})^{-1} &= a \\
 (a * b)^{-1} &= b^{-1} * a^{-1} \\
 a * x = b &\implies x = a^{-1} * b \\
 x * a = b &\implies x = b * a^{-1} \\
 a^n &= \underbrace{a * a * \cdots * a}_{n \text{ times}} \\
 a^0 &= e, \quad a^{-n} = (a^{-1})^n \\
 a^{m+n} &= a^m * a^n, \quad (a^m)^n = a^{mn}
 \end{aligned}$$

— **Quotient group / Groupe quotient :**

$$\begin{aligned}
 H &\text{ subgroup of } G \\
 a \sim b &\iff a * b^{-1} \in H \\
 G/H &= \{ \dot{a} \mid a \in G \} \\
 \dot{a} * \dot{b} &= \dot{a * b} \\
 \text{If } G &\text{ abelian, then } G/H \text{ is abelian group}
 \end{aligned}$$

— **Group homomorphism / Homomorphisme de groupes :**

$$f : G \rightarrow H, \quad \forall a, b \in G, f(a * b) = f(a) \star f(b)$$

$$\begin{aligned}
 f(e_G) &= e_H \\
 f(a^{-1}) &= (f(a))^{-1} \\
 \text{Im}(f) &= f(G) \text{ subgroup of } H \\
 \text{Ker}(f) &= f^{-1}(e_H) \text{ subgroup of } G \\
 f \text{ injective} &\iff \text{Ker}(f) = \{e_G\} \\
 f \text{ surjective} &\iff \text{Im}(f) = H
 \end{aligned}$$

— **Isomorphism / Isomorphisme :** Bijective homomorphism

$$\begin{aligned}
 G &\cong H \text{ if } \exists f : G \rightarrow H \text{ isomorphism} \\
 f^{-1} &\text{ is also isomorphism} \\
 \text{If } f &\text{ isomorphism, then } |G| = |H|
 \end{aligned}$$

— **Endomorphism / Endomorphisme :** $f : G \rightarrow G$ homomorphism

— **Automorphism / Automorphisme :** Bijective endomorphism

4.3 Rings / Anneaux

— **Definition / Définition :** $(A, +, \cdot)$ is a ring if :

1. $(A, +)$ is abelian group (neutral 0)
2. $\cdot$ is associative
3. $\cdot$ is distributive over $+$:

$$\begin{aligned}
 a \cdot (b + c) &= a \cdot b + a \cdot c \\
 (b + c) \cdot a &= b \cdot a + c \cdot a
 \end{aligned}$$

— **Commutative ring / Anneau commutatif :** $\cdot$ is commutative

- **Unitary ring / Anneau unitaire** : $\exists 1 \in A, \forall a \in A, 1 \cdot a = a \cdot 1 = a$
- **Zero divisor / Diviseur de zéro** :

$$\exists a, b \neq 0, a \cdot b = 0$$

- **Integral domain / Anneau intègre** : Commutative, unitary, no zero divisors

$$a \cdot b = 0 \implies a = 0 \text{ or } b = 0$$

- **Ring properties / Propriétés des anneaux** :

$$0 \cdot a = a \cdot 0 = 0$$

$$a \cdot (-b) = (-a) \cdot b = -(a \cdot b)$$

$$(-a) \cdot (-b) = a \cdot b$$

$$a \cdot (b - c) = a \cdot b - a \cdot c$$

$$(b - c) \cdot a = b \cdot a - c \cdot a$$

$$\sum_{i=1}^m a_i \cdot \sum_{j=1}^n b_j = \sum_{i=1}^m \sum_{j=1}^n a_i \cdot b_j$$

- **Subring / Sous-anneau** : $B \subseteq A$ is subring if :

$$B \neq \emptyset$$

$$\forall a, b \in B, a - b \in B$$

$$\forall a, b \in B, a \cdot b \in B$$

If A unitary and $1 \in B$, then B is unitary subring

- **Ideal / Idéal** : $I \subseteq A$ is ideal if :

1. I is subgroup of $(A, +)$
2. $\forall a \in A, \forall i \in I, a \cdot i \in I$ (left ideal)
3. $\forall a \in A, \forall i \in I, i \cdot a \in I$ (right ideal)

Two-sided ideal / Idéal bilatère : Both left and right ideal

- **Principal ideal / Idéal principal** : $I = aA = \{a \cdot x \mid x \in A\}$
- **Ring homomorphism / Homomorphisme d'anneaux** :

$$f : A \rightarrow B, \quad \forall a, b \in A$$

$$f(a + b) = f(a) + f(b)$$

$$f(a \cdot b) = f(a) \cdot f(b)$$

If A, B unitary : $f(1_A) = 1_B$ (unitary homomorphism)

$$f \text{ injective} \iff \text{Ker}(f) = \{0\}$$

$$f \text{ surjective} \implies f(1_A) = 1_B$$

- **Quotient ring / Anneau quotient** :

I ideal of A

$$A/I = \{\dot{a} \mid a \in A\}$$

$$\dot{a} + \dot{b} = \dot{a + b}$$

$$\dot{a} \cdot \dot{b} = \dot{a \cdot b}$$

If A commutative, then A/I commutative

If A unitary, then A/I unitary with $\dot{1}$

— **Examples / Exemples :**

$(\mathbb{Z}, +, \cdot)$: integral domain

$(\mathbb{Z}/n\mathbb{Z}, +, \cdot)$: ring

$\mathbb{Z}/n\mathbb{Z}$ is field iff n prime

$n\mathbb{Z}$ ideal of $\mathbb{Z}$

4.4 Fields / Corps

— **Definition / Définition :** $(\mathbb{K}, +, \cdot)$ is a field if :

1. $(\mathbb{K}, +, \cdot)$ is commutative unitary ring
2. $\mathbb{K}^* = \mathbb{K} \setminus \{0\}$ is group under $\cdot$

— **Field properties / Propriétés des corps :**

Every field is integral domain

$$a \cdot b = 0 \implies a = 0 \text{ or } b = 0$$

$$a \cdot b = a \cdot c, a \neq 0 \implies b = c$$

The only ideals are $\{0\}$ and $\mathbb{K}$

— **Subfield / Sous-corps :** $\mathbb{K}' \subseteq \mathbb{K}$ is subfield if :

$$\mathbb{K}' \neq \emptyset$$

$$\forall a, b \in \mathbb{K}', a - b \in \mathbb{K}'$$

$$\forall a, b \in \mathbb{K}', b \neq 0, a \cdot b^{-1} \in \mathbb{K}'$$

— **Characteristic / Caractéristique :**

$\text{char}(\mathbb{K}) = \text{smallest } n > 0 \text{ such that } n \cdot 1 = 0$

If no such n , $\text{char}(\mathbb{K}) = 0$

$\text{char}(\mathbb{K})$ is prime or 0

$$\text{char}(\mathbb{Z}/p\mathbb{Z}) = p \quad (p \text{ prime})$$

— **Examples / Exemples :**

$$(\mathbb{Q}, +, \cdot), (\mathbb{R}, +, \cdot), (\mathbb{C}, +, \cdot)$$

$$(\mathbb{Z}/p\mathbb{Z}, +, \cdot) \text{ for } p \text{ prime}$$

Part 5 : Important Formulas / Formules importantes

5.1 Counting Formulas / Formules de dénombrement

— **Cardinality formulas / Formules de cardinalité :**

If $A \cap B = \emptyset$, then $|A \cup B| = |A| + |B|$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$|A \times B| = |A| \times |B|$$

$$|\mathcal{P}(A)| = 2^{|A|}$$

$$|A^B| = |A|^{|B|} \quad (\text{functions from } B \text{ to } A)$$

— **Inclusion-Exclusion Principle / Principe d'inclusion-exclusion :**

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

5.2 Algebraic Identities / Identités algébriques

— For groups / Pour les groupes :

$$(a * b)^n = a^n * b^n \quad (\text{if commutative})$$

$$(a^{-1})^n = (a^n)^{-1} = a^{-n}$$

$$a^m * a^n = a^{m+n}$$

$$(a^m)^n = a^{mn}$$

— For rings / Pour les anneaux :

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (\text{if commutative})$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

— Binomial theorem / Formule du binôme :

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

5.3 Properties Tables / Tables de propriétés

Structure	Associative	Commutative	Neutral	Inverse	Distributive
Group	Yes	(Abelian)	Yes	Yes	No
Ring (+)	Yes	Yes	0	Yes	
Ring (·)	Yes	(Commutative)	1	(Units only)	Over +
Field (+)	Yes	Yes	0	Yes	
Field (·)	Yes	Yes	1	Yes (except 0)	Over +